

Sublevel set probabilities of the CATE function

Anders Munch
joint work with Thomas Gerds

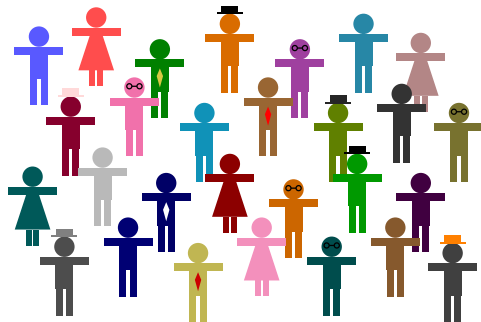
Section of Biostatistics
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Reporting heterogeneity – a trade off



ATE



CATE

Some notation

$Y^a \in \{0, 1\}$ potential outcomes under different treatment levels $a \in \{0, 1\}$

$W \in \mathbb{R}^d$ individual baseline characteristics

A actually given treatment

Y actually observed outcome

The CATE function (with respect to W) is

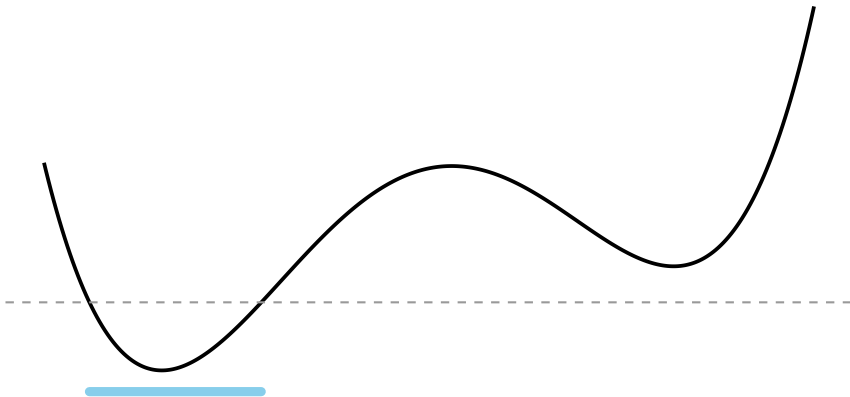
$$w \longmapsto \tau(w) = \mathbb{E}[Y^1 - Y^0 \mid W = w].$$

It can be identified under standard causal assumptions as

$$\tau(w) = \mathbb{E}[Y \mid A = 1, W = w] - \mathbb{E}[Y \mid A = 0, W = w].$$

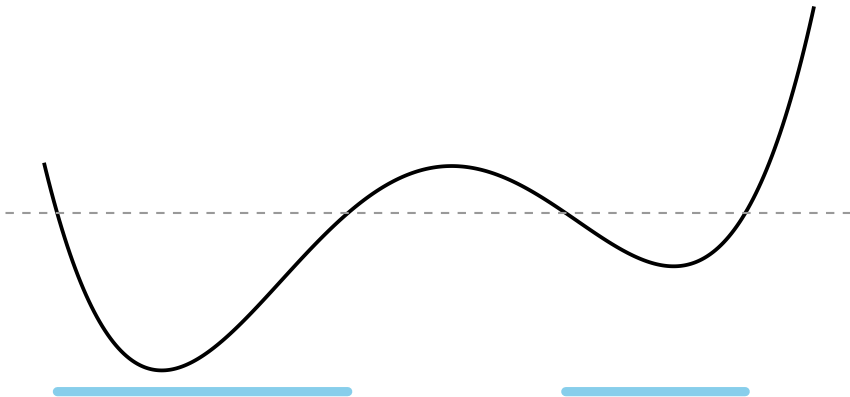
Sublevel sets of the CATE function τ

$$\mathcal{W}_\alpha = \{w \in W : \tau(w) \leq \alpha\}, \quad \alpha \in [-1, 1]$$



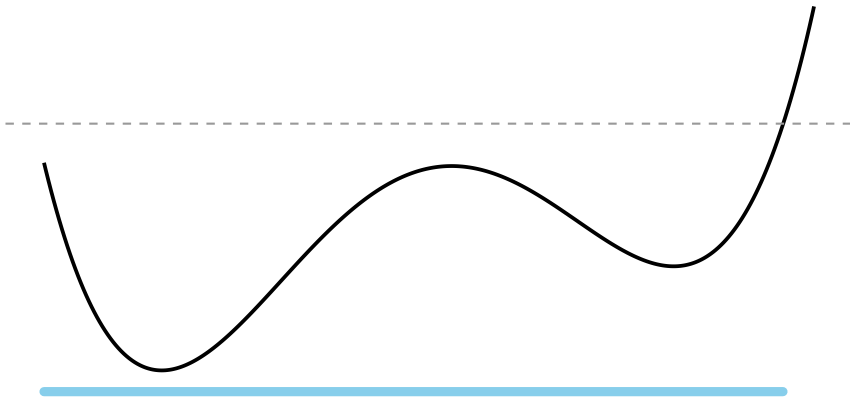
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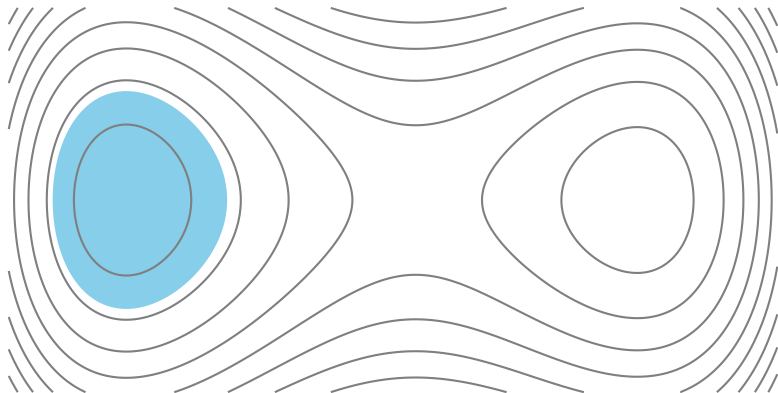
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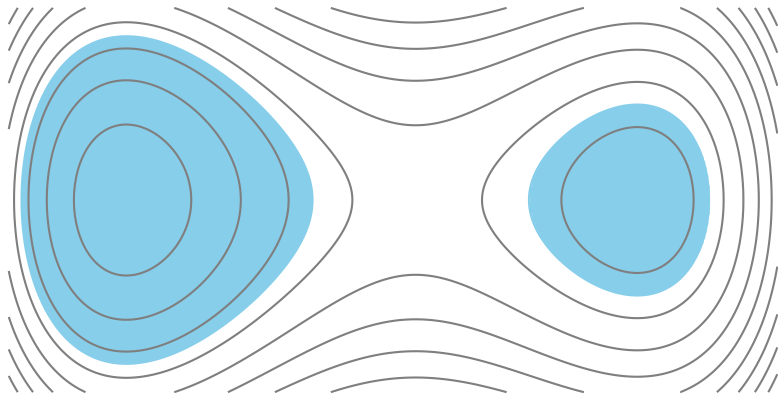
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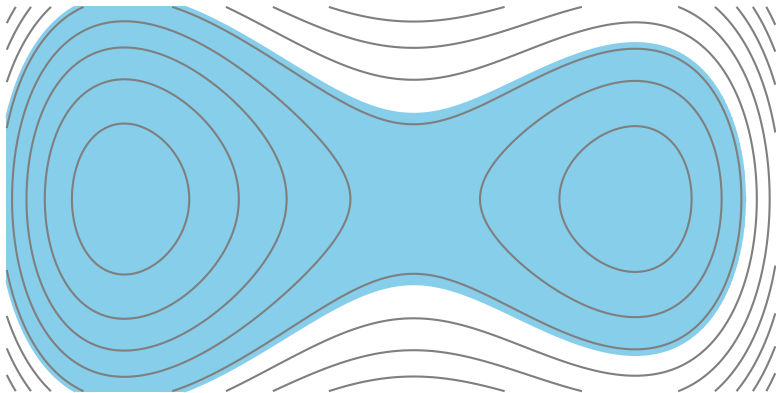
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Sublevel set size function

$$\gamma: [-1, 1] \longrightarrow [0, 1], \quad \gamma(\alpha) = P(\mathcal{W}_\alpha) = P(\tau(W) \leq \alpha).$$

CDF of the random variable $\tau(W)^\dagger$, i.e., a **univariate, nondecreasing function**.

The value $\gamma(\alpha)$ is the proportion of the population with an expected treatment effect (given available baseline information W) below or equal to α . E.g., $\gamma(0)$ is the proportion of the population for which the treatment is not expected to be beneficial, given available baseline information.

[†]Closely related to “the sorted effects method” [Chernozhukov et al., 2018b]

Illustration of the function γ

`glm(Y ~ age * A + sex * A)`

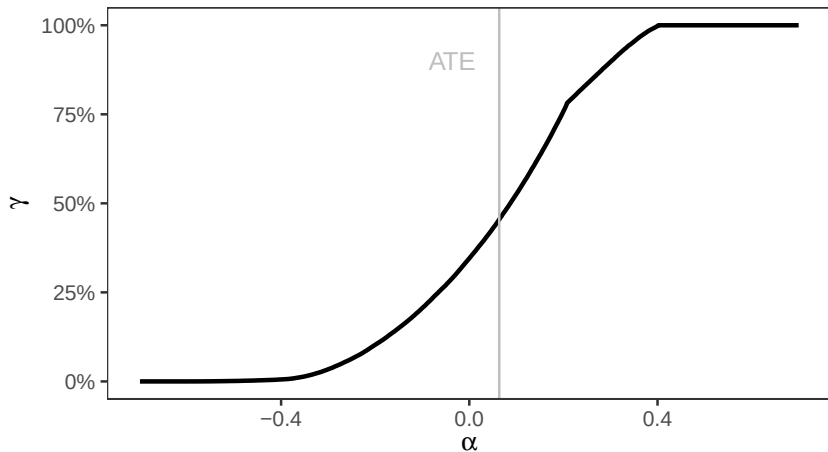


Illustration of the function γ

`glm(Y ~ age + sex + A)`

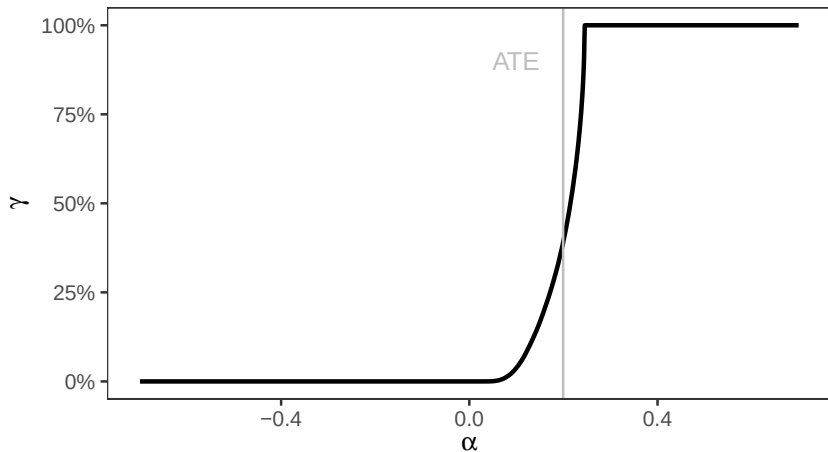
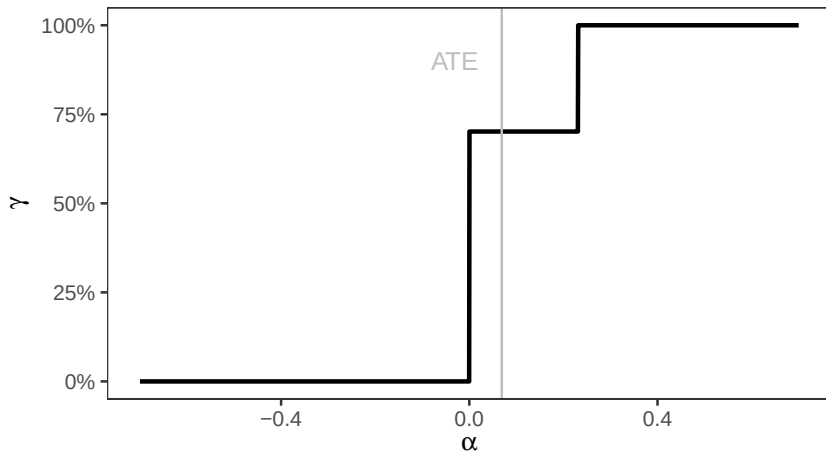


Illustration of the function γ

`glm(Y ~ sex * A)`



Estimation

Estimation of γ is “non-standard”

One possible estimator is the plug-in estimator

$$\hat{\gamma}_n(\alpha) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{\hat{\tau}_n(W_i) \leq \alpha\},$$

based on some CATE learner $\hat{\tau}_n$. Not easy to analyze its asymptotic behaviour when data-adaptive methods are used.

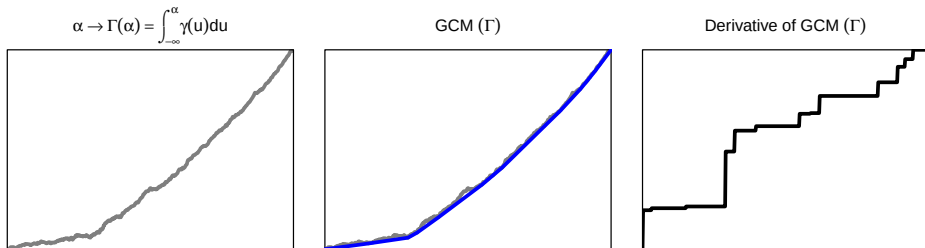
Efficient influence function based methods like targeted/debiased machine learning address this.[†]

Complicated by the fact that the parameter $\gamma(\alpha)$ is not pathwise differentiable.

[†]Hines et al. [2022], Chernozhukov et al. [2018a], van der Laan and Rose [2011]

Generalized Grenander-type estimator[†]

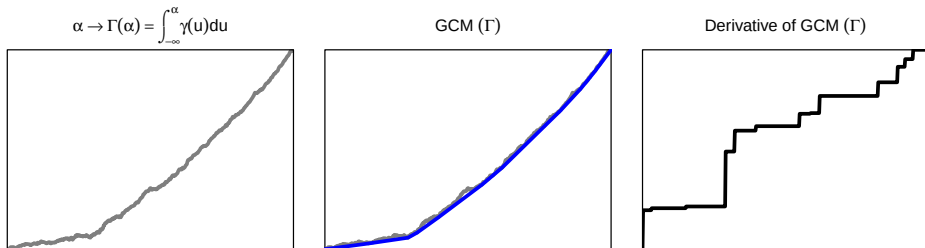
The primitive of a nondecreasing function is convex. Suggests the estimation strategy:



[†]Westling and Carone [2020], van der Vaart and van der Laan [2006], Groeneboom [1983], Grenander [1956]

Generalized Grenander-type estimator[†]

The primitive of a nondecreasing function is convex. Suggests the estimation strategy:



The parameter $\Gamma(\alpha)$ *is* pathwise differentiable at values α at which γ is continuous.

—→ Standard toolbox available for estimating Γ .

[†]Westling and Carone [2020], van der Vaart and van der Laan [2006], Groeneboom [1983], Grenander [1956]

Estimation of Γ

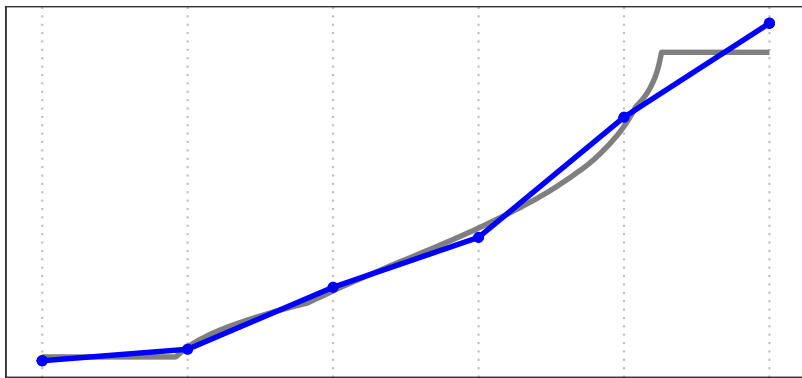
At values α at which γ is continuous, we can find the efficient influence function $\varphi(\alpha)$ of $\Gamma(\alpha)$ and use the one-step estimator

$$\hat{\Gamma}_n^*(\alpha) = \hat{\Gamma}_n(\alpha) + \frac{1}{n} \sum_{i=1}^n \hat{\varphi}_n(\alpha).$$

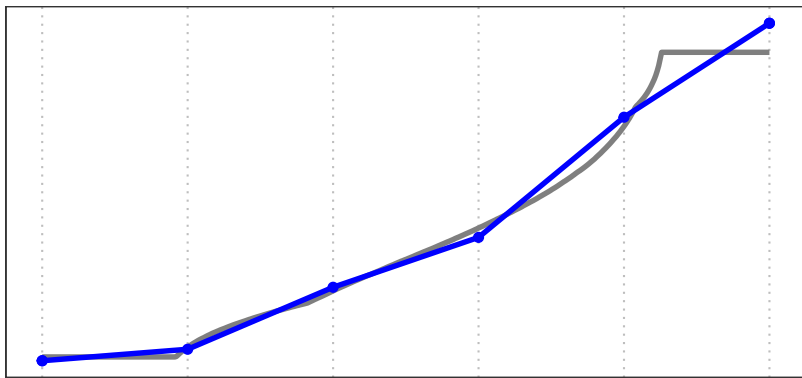
Allows for the use of flexible, non-parametric estimation techniques for estimating the nuisance parameters in Γ and φ .

Westling and Carone [2020] provide result for the asymptotic distribution of the derivative of $\text{GCM}(\hat{\Gamma}_n^*)$ under some conditions on $\hat{\Gamma}_n^*$.

Spline approximation of γ



Spline approximation of γ



This **curve** is determined by coefficients that can be written as linear functionals of Γ .

→ Standard toolbox available again.

Estimation of best piece-wise linear approximation

The best piece-wise linear $\mathcal{L}^2$ -approximation of γ over $a_1, \dots, a_L$ is the function

$$\gamma^\#(\alpha) = \sum_{l=1}^L H_l(\alpha) \gamma_l^\#, \quad \text{where} \quad \gamma_l^\# = [M^{-1}\zeta]_l,$$

H_l are the hat functions, and the coefficient are

$$M^{-1}\zeta, \quad \text{where} \quad M_{lj} = \langle H_l, H_j \rangle \quad \text{and} \quad \zeta_l = \langle H_l, \gamma \rangle, \quad \text{for } l, j = 1, \dots, L.$$

We have

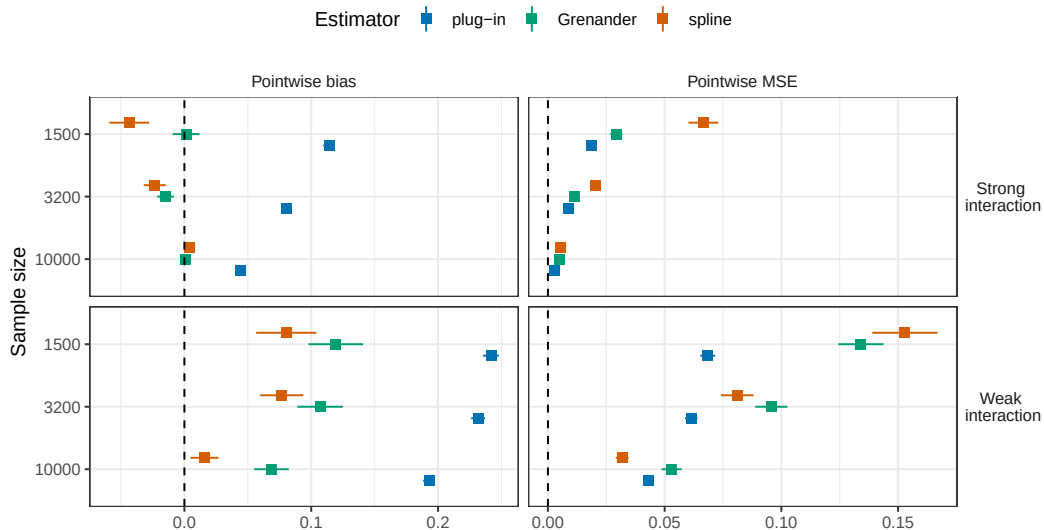
$$\zeta_1 = -\Gamma(a_1) + \frac{1}{a_2 - a_1} \int_{a_1}^{a_2} \Gamma(u) \, du, \quad \dots$$

and ζ has efficient influence function

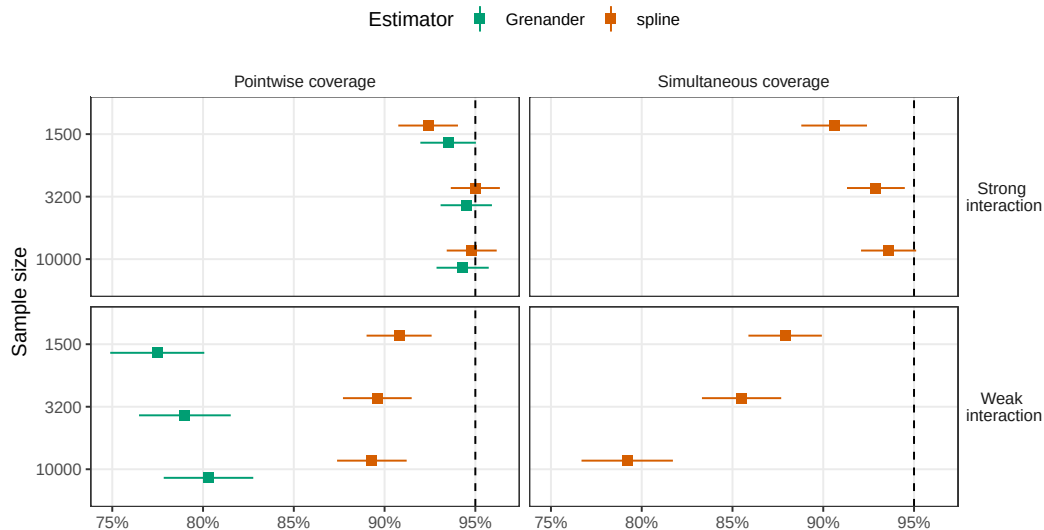
$$-\varphi(a_1) + \frac{1}{a_2 - a_1} \int_{a_1}^{a_2} \varphi(u) \, du, \quad \dots$$

Joint asymptotic distribution for $\hat{\zeta}_n^\star$ allows for construction of confidence bands.

Results from some numerical experiments



Results from some numerical experiments



Recap and outlook

Univariate monotone function for visualizing and summarizing treatment heterogeneity.

“Non-standard” estimation problem

- Monotone function estimation techniques
- Approximate spline estimand

A spline approximation of a monotone function needs not be monotone itself.

Our suggested estimators work under a continuity assumption.

Estimating the sublevel set size function γ still seems to be a statistically challenging problem in a nonparametric model.

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Investigating treatment heterogeneity through sublevel-set probabilities

Anders Munch and Thomas Gerds

available on arXiv soon